

Malhar Gudekar

(217) -766-0681 — gudekar2@illinois.edu — LinkedIn — Portfolio — GitHub

EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

- **Machine Learning / Statistics:** Classification, Regression, Feature Engineering, Model Evaluation, Anomaly Detection, Precision, Recall, Accuracy, Statistical Analysis, Predictive Analytics, Data Validation
- **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j, Scikit-learn, LightGBM
- **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization
- **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

- Designed and implemented end-to-end **machine learning** pipeline to ingest, process, and analyze wearable data from 100+ users, combining scalable **data ingestion**, **feature engineering**, and model-ready transformations to enable longitudinal behavioral insights and downstream analytics
- Developed and optimized **PostgreSQL** data models, indexing strategies, and query execution plans for high-volume datasets, improving query performance by **38%** while enabling efficient feature extraction for machine learning workflows
- Built embedding-based retrieval and summarization pipeline to enhance contextual understanding of user interactions, improving multi-turn response accuracy by 43% and supporting more reliable behavioral analysis

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Developed **NLP**-based machine learning pipelines to detect health misinformation from large-scale social media data, applying text preprocessing, **feature extraction**, and **optimization** techniques to reduce processing latency by **41%**
- Built distributed **data pipelines** using **Kafka** and **Neo4j** Streams to enable **scalable ingestion**, transformation, and real-time analysis of high-volume unstructured datasets
- Designed production-grade inference APIs and deployment workflows to support observability, predictive health monitoring, and consistent evaluation of AI/ML model performance outputs

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

- Built and deployed **Retrieval-Augmented Generation (RAG)** system using embeddings and **vector search**, combining data pipelines, feature engineering, and retrieval optimization to improve document retrieval efficiency by 64%
- Designed scalable ETL and log aggregation pipelines using FastAPI and PostgreSQL to transform unstructured system data into structured formats for predictive analytics and machine learning
- Optimized indexing strategies and data processing workflows to enable low-latency querying and scalable deployment of AI-driven systems

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

- Applied anomaly detection techniques to large-scale operational system logs using Python and PySpark to identify trends and performance issues, combining exploratory analysis and root cause investigation to surface actionable insights
- Built Tableau and Power BI dashboards to visualize key performance indicators and translate operational metrics into clear data narratives for stakeholders, reducing reporting time by 40%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

- Developed and maintained Power BI dashboards using **SQL** and **DAX**, transforming raw data into actionable insights for business decision-making across multiple domains.
- Performed data validation, cleaning, and transformation on large multi-source datasets using **SQL** and **Excel**, improving data accuracy by over 30% and reducing reporting inconsistencies, enabling faster and more reliable decision-making across sales, finance, and operations teams

Stu/Dio at Illinois

August 2025 - Present

Project Manager

- Led a **cross-functional** team of six to deliver and scale Deepcover game enhancements by leveraging user data, performance metrics, and feedback analysis to drive feature prioritization and improve games outcomes

PROJECTS

Scalable ML System for Customer Risk Prediction

- Built **LightGBM**-based machine learning model to **predict customer credit risk** from **time-series transaction data**, combining feature engineering, classification modeling, and evaluation techniques to identify high-risk and low-risk segments.
- Performed feature engineering on behavioral patterns such as transaction frequency and spending trends, improving **model precision** by 20% while enhancing robustness and interpretability.
- Evaluated **model performance** using precision, recall, and accuracy, and designed a decision framework to translate predictions into actionable credit limit recommendations.